

RGGE-DTD: A Unified Model for Simultaneous Prediction of Drug-Target Interactions and Drug-Disease Associations in Drug Repositioning

Junrong Song^{a,b,1,*}, Zilong Yang^a, Zhiming Song^{a,b}, Yan Heng^a, Lichang Ge^a, Jinpeng Zhang^a, Yajuan Chen^{c,*}

^a The School of Information, Yunnan University of Finance and Economics, Kunming, Yunnan 650221, China

^b Yunnan Key Laboratory of Service Computing, Yunnan University of Finance and Economics, Kunming, Yunnan 650221, China

^c Yunnan Key Laboratory of Breast Cancer Precision Medicine, Academy of Biomedical Engineering, Kunming Medical University, Kunming, China

ARTICLE INFO

Keywords:

Heterogeneous network
Drug-target interaction
Drug-disease association
Relational Graph Convolutional Network
Gelato
Class imbalance

ABSTRACT

In the field of drug discovery, accurate prediction of drug-target interactions (DTIs) and drug-disease associations (DDAs) is pivotal for effective drug repositioning. However, most existing methods model these tasks in isolation, neglecting their intrinsic coupling, and suffer from noisy heterogeneous networks, severe class imbalance, and inadequate modeling of relation heterogeneity. To address these challenges, we propose RGGE-DTD, a unified framework that jointly predicts DTIs and DDAs by integrating relational graph convolution with autocovariance-based graph enhancement. The model leverages an enhanced heterogeneous network of drugs, targets, and diseases to enable relation-aware, cross-task representation learning and employs a composite loss function combining weighted cross-entropy and N-pair contrastive loss to mitigate bias from class imbalance. Extensive experiments show that RGGE-DTD consistently outperforms ten state-of-the-art methods on both DTI and DDA prediction tasks, achieving higher AUPR scores under rigorous unbiased evaluation settings. Case studies further validate its capacity to identify high-confidence, biologically plausible associations not present in current knowledge bases, highlighting its potential to support reliable computational drug repositioning.

1. Introduction

With the rising demand for new therapeutic options and the high costs associated with de novo drug development, drug repositioning has emerged as a promising strategy to accelerate treatment discovery (Norouzi, Norouzi, Abbasi, Norouzi, & Razzaghi, 2025). Identifying drug-target(i.e., protein) interactions (DTIs) and characterizing drug-disease associations (DDAs) are two intertwined tasks in drug repositioning (B. Wang, et al., 2023). Drugs exert therapeutic effects primarily by modulating the expression or activity of target proteins—through upregulation, downregulation, or functional inhibition—enabling their potential application across diverse disease contexts (Lian, Wang, & Du, 2023). However, due to the poly-pharmacological nature of drugs, many targets and indications remain undiscovered (Santos, et al., 2017). While experimental validation of these relationships is often accurate, it is typically time-consuming and costly (Shi & Yiu, 2015). In contrast, in silico prediction offers a scalable

and efficient means to infer potential relationships among drugs, targets, and diseases (Masoudi-Nejad, Mousavian, & Bozorgmehr, 2013). Yet, most existing methods treat DTI and DDA prediction as isolated tasks, constrained by data heterogeneity, methodological divergence, and architectural limitations. This compartmentalization impedes a holistic understanding of the mechanistic interplay between drug actions and disease modulation, limiting the translational potential of computational drug repositioning.

Traditional computational approaches to drug repositioning have primarily focused on identifying interactions between new drugs and known targets, or between existing drugs and novel targets. These methods fall into three main categories: docking-based, ligand-based, and similarity-based approaches. Docking-based methods require high-resolution three-dimensional structures of target proteins to estimate binding affinities, but their applicability is limited by the scarcity of experimentally determined structures (Ballester & Mitchell, 2010; Morris, et al., 2009; Shaikh, Sharma, & Garg, 2016; Śledź & Caflisch,

* Corresponding author.

E-mail addresses: zz2144@ynufe.edu.cn (J. Song), 202302110732@stu.ynufe.edu.cn (Z. Yang), zz2145@ynufe.edu.cn (Z. Song), 202402110821@stu.ynufe.edu.cn (Y. Heng), 202302110730@stu.ynufe.edu.cn (L. Ge), zjp@ynufe.edu.cn (J. Zhang), chenyajuan@kmmu.edu.cn (Y. Chen).

¹ 0009-0000-6716-5975

<https://doi.org/10.1016/j.eswa.2026.132319>

Received 14 September 2025; Received in revised form 11 March 2026; Accepted 31 March 2026

Available online 4 April 2026

0957-4174/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

2018). Ligand-based methods leverage known active ligands of a target protein to build predictive models; however, their performance declines sharply when only a small number of known active ligands are available (Hansch, Maloney, Fujita, & Muir, 1962; Keiser, et al., 2007). Similarity-based methods operate under the “guilt-by-association” principle—chemically similar drugs are likely to interact with the same targets, and functionally related targets tend to bind similar ligands (He, Heide-meyer, Ban, Cherkasov, & Ester, 2017; Öztürk, Özgür, & Ozkirimli, 2018; Xia, Wu, Zhou, & Wong, 2010). These methods typically compute drug and target similarities independently from chemical structures and protein sequences, then train classifiers to predict interactions based on these features. While integrating multiple similarity metrics often improves accuracy over single-view models, such heuristic fusion can introduce noise or irrelevant information, degrading model robustness (Bagherian, et al., 2021; Lian, et al., 2023).

Computational drug-disease association prediction is another cornerstone of drug repositioning. Current approaches can be broadly classified into matrix-based methods, traditional machine learning, and deep learning. Matrix-based methods typically address the sparsity of DDA matrices through low-rank approximation for matrix completion (H. Luo, et al., 2018; Y. Luo, et al., 2017; M. Yang, Luo, Li, & Wang, 2019) or decompose the observed interaction matrix into latent factors representations whose product approximates the origin matrix (F. Huang, Qiu, Li, Liu, & Ni, 2020; Sadeghi, Lu, & Ngom, 2022; M. Yang, Wu, Zhao, Li, & Wang, 2021). While effective in capturing global patterns, these methods rely heavily on existing annotations and often struggle with limited labels or complex nonlinear relationships. Machine-learning-based approaches are widely adopted due to their strong predictive performance (Gottlieb, Stein, Rupp, & Sharan, 2011; Kitsiranuwat, Suratanee, & Plaimas, 2022; K. Yang, Zhao, Waxman, & Zhao, 2019). These methods typically involve handcrafted feature engineering—encoding drugs and diseases into fixed-dimensional vectors based on chemical, genomic, or phenotypic profiles—followed by classifier training. However, this two-stage pipeline depends on domain expertise for feature selection, limiting its generalizability and scalability in real-world applications. Deep learning models offer a promising alternative by enabling automatic feature extraction and end-to-end optimization (Li, et al., 2020; Norouzi, et al., 2025; Peng, Li, & Shang, 2020; Xuan, Ye, Zhang, Zhao, & Sun, 2019). By learning hierarchical representations directly from raw or semi-processed inputs, these models mitigate the need for manual feature design and have demonstrated superior performance in various biomedical prediction tasks.

In recent years, network-based approaches have emerged as a powerful paradigm for drug repositioning by modeling the complex relationships among drugs, targets, and diseases within a heterogeneous interaction network (HIN). These methods integrate diverse biological entities into unified graph structures, enabling the propagation of information across modalities and yielding superior performance in both drug–target interaction (DTI) and drug–disease association (DDA) prediction (M. Yang, et al., 2021; Zhao, Hu, You, Wang, & Su, 2022). For DTI prediction, several graph neural network (GNN)-based models have been developed. NeoDTI leverages neural message passing to learn topology-aware embeddings that preserve structural proximity in the network (Wan, Hong, Xiao, Jiang, & Zeng, 2019). EEG-DTI utilizes graph convolutional networks (GCNs) to extract low-dimensional feature representations of drugs and targets (Peng, et al., 2021), while AMGDTI enhances this framework by adaptively searching a meta-graph to guide multi-hop message aggregation across heterogeneous biomedical networks (Su, et al., 2024). Similarly, DDA prediction has benefited from advances in geometric deep learning. DeepDR fuses ten heterogeneous networks via a deep autoencoder to learn drug embeddings and infers novel indications using a variational autoencoder (Zeng, et al., 2019). LAGCN employs GCNs with an attention mechanism to combine hierarchical node representations across convolutional layers (Yu, Huang, Zhao, Xiao, & Zhang, 2021). DDAGDL employs an attention-

based geometric deep learning method to generate smoothed embeddings from HINs and uses an XGBoost for final prediction (Zhao, Su, et al., 2022). Additionally, the HINGRL model integrates HIN using autoencoders, learns network representations via DeepWalk, and employs a Random Forest classifier to predict potential drug–disease associations (R. Yang, Fu, Zhang, & Zhang, 2024).

Despite these advances, current network-based methods still face several critical limitations. First, many rely on predefined or manually curated HIN, which may contain noisy, incomplete or biased annotations—compromising the quality of learned representations. Second, they typically treat all edge types equally, neglecting the varying biological significance of different relational types (e.g., drug–target binding vs. disease comorbidity), thereby limiting the model’s ability to capture nuanced interactions. Third, severe class imbalance—due to the scarcity of experimentally validated positive associations—commonly leads to negative sampling strategies that discard potentially informative unlabeled pairs, introducing training bias and inflating performance estimates (Wan, et al., 2019; B. Wang, et al., 2023; Zeng, et al., 2019). Most critically, the vast majority of existing models address DTI and DDA prediction as isolated tasks, despite the intrinsic interdependence among drugs, targets, and diseases in pharmacological contexts (Tanvir et al., 2023). To our knowledge, only HeTriNet (Tanvir, et al., 2023) and DTD-MTL (B. Wang, et al., 2023) attempt joint prediction; however, both remain constrained by suboptimal network construction, lack of relation-aware message weighting, and unbiased training protocols. This gap underscores the need for unified, robust, and end-to-end frameworks.

To address these limitations, we develop RGGE-DTD that integrates relational graph learning (RGCN) (Schlichtkrull, et al., 2018), autocovariance-based graph enhancement (Gelato) (Huang et al., 2023), and uncertainty-aware multi-task optimization to jointly model DTIs and DDAs. Unlike existing approaches that treat these tasks in isolation or rely on rigid graph constructions (Abbasi, et al., 2025), our model enables co-adaptive representation learning across drugs, targets, and diseases, thereby improving generalization for both well-studied and novel entities. Experimental results demonstrate that RGGE-DTD consistently outperforms ten state-of-the-art methods on both DTI and DDA prediction tasks, offering a robust and biologically plausible foundation for computational drug repositioning.

The main contributions of the proposed approaches are as follows: (1) A unified heterogeneous graph learning framework that combines RGCN and Gelato to jointly leverage node attributes and high-order topological dependencies for multi-task prediction; (2) An adapted graph enhancement strategy that fuses molecular features with network topology via autocovariance scoring, improving resilience to structural sparsity and noise; (3) An unbiased multi-task training protocol that combines N-pair loss and weighted cross-entropy under stratified negative sampling, mitigating class imbalance while preserving the natural distribution of unknown interactions.

2. Methodology

2.1. Overview

RGGE-DTD is a unified framework for the joint prediction of drug–target interactions (DTIs) and drug–disease associations (DDAs). As illustrated in Fig. 1, the method starts by constructing a heterogeneous information network (HIN) by integrating six types of biomedical associations: drug–protein interactions (DP), drug–disease associations (DS), drug–drug interactions (DD), protein–disease associations (PS), protein–protein interactions (PP), and disease–disease associations (SS), with preprocessed molecular features assigned as node attributes. Next, RGCN is applied to learn relation-specific node embeddings by aggregating neighborhood information across different edge types in the HIN. To enhance structural robustness, Gelato is employed to refine the graph topology by concatenating the learned embeddings with node attributes

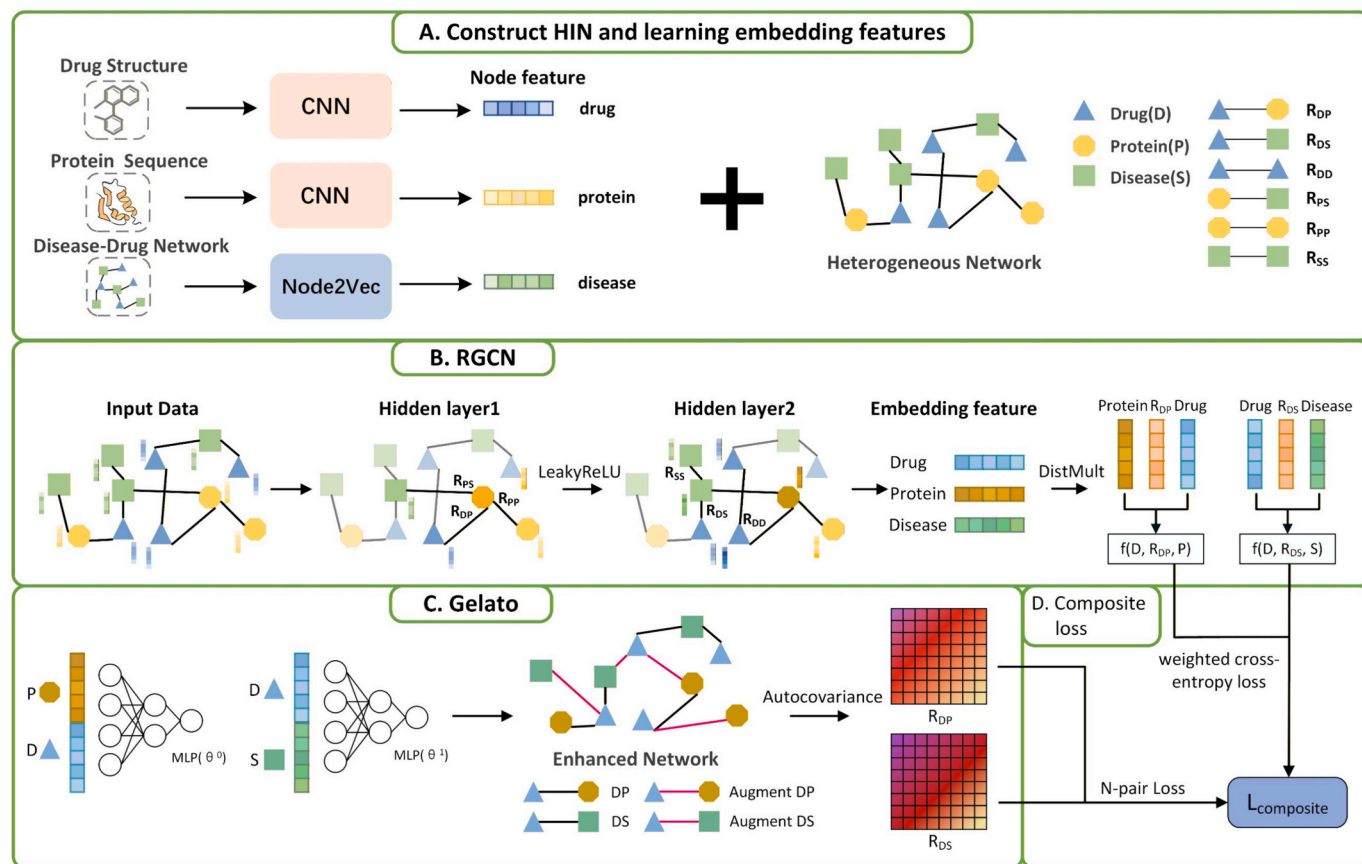


Fig. 1. The overall architecture of RGGE-DTD (A) Construction of HIN and learning of biological embedding features. (B) Node representation learning via RGCN. (C) Gelato for predicted link prediction. (D) Composite loss function to optimize the model.

and computing an autocovariance (AC)-based pairwise score matrix to capture high-order topological dependencies. Finally, the model is optimized using an unbiased multi-task learning framework with a composite loss function that combines N-pair loss and weighted cross-entropy loss. The resulting embeddings are used to simultaneously predict DTIs and DDAs.

2.2. Datasets

In this study, we evaluated our proposed model using Luo's database (Y. Luo, et al., 2017). From this dataset, we extracted three types of nodes (drugs, proteins, and diseases) and five types of edges (drug-drug, drug-disease, drug-protein, protein-protein, and protein-disease). The chemical structures of drugs, denoted as SMILES, were obtained from the DrugBank database (version 3.0) (Knox, et al., 2010) and encoded into integers with a maximum length of 100 characters to cover at least 90% of the compounds (Öztürk, et al., 2018). These SMILES sequences were used as the initial node features for drugs. Similarly, protein sequences (denoted as FASTA) were extracted from the HPRD database (version 9) (Keshava Prasad, et al., 2009) and encoded into integer sequences with a maximum length of 1000 to cover at least 85% of the proteins, serving as the initial node features for proteins. Drug-disease interaction networks were used as initial features for disease nodes and were acquired from the Comparative Toxicogenomics Database (Davis, et al., 2012). Known drug-protein interactions and drug-drug interactions were obtained from DrugBank (version 3.0) (Knox, et al., 2010), protein-protein interactions from HPRD (version 9) (Keshava Prasad, et al., 2009), and drug-disease and protein-disease associations from the Comparative Toxicogenomics Database (Davis, et al., 2012). The disease-disease similarity matrix was constructed based on shared drug and protein associations (see Section 2.4 for details). An overview

of the network statistics is provided in Table 1. The network densities were calculated accordingly: 5.02% for drug-disease associations and 0.18% for drug-protein interactions. These low densities indicate that the networks are highly sparse and exhibit a significant imbalance between positive and negative samples.

2.3. Encoding the node representations by CNN and Node2Vec

Drug, protein, and disease features must first be extracted and transformed into representative vector embeddings prior to being fed into the RGCN. For drugs and proteins, we adopted a sequence-based feature extraction framework. Specifically, SMILES strings and amino acid sequences were tokenized and mapped to integer sequences using predefined vocabularies, resulting in encoded input Seq_{drug} and $Seq_{protein}$, respectively. To capture local structural motifs and hierarchical patterns in these sequences, we employed a one-dimensional convolutional neural network (1D CNN), which has demonstrated effectiveness in molecular representation learning (Dehghan, Razzaghi, Abbasi, &

Table 1
The detailed description of our dataset

Edge/Node-type	Num	Density(%)
Drug-Drug interaction	10036	2.00
Drug-Protein interaction	1923	0.18
Drug-Disease association	199214	5.02
Protein-Protein interaction	7363	0.32
Protein-Disease association	1596745	18.84
Disease-Disease interaction	4344310	13.84
Drug	708	—
Protein	1512	—
Disease	5603	—

Gharaghani, 2023). This architecture includes convolutional and pooling layers (Kiranyaz, Ince, Abdeljaber, Avci, & Gabbouj, 2019). The convolutional layers scan the input to detect local patterns, while the pooling layers reduce dimensionality and increase the receptive field. The processed outputs are denoted as drug feature embeddings X_D and protein feature embeddings X_P .

For disease nodes, which lack intrinsic sequences, we derived semantic features from the drug-disease interaction network using Node2Vec (Grover & Leskovec, 2016). This approach performs biased random walks to generate node sequences that preserve network topology, followed by training low-dimensional embeddings using the Skip-gram model. The resulting disease feature vectors are denoted as X_S . This strategy enables the integration of higher-order neighborhood information into the node representations, which is critical for downstream graph learning tasks. Similar approaches have been applied in biomedical knowledge graphs (Su, et al., 2024). All extracted features (X_D , X_P , X_S) are then projected as initial node embeddings for input to the RGCN.

2.4. Constructing heterogeneous networks

Utilizing known biological association data, we constructed a heterogeneous network denoted as $G = (V, E)$, where V represents the node set and E denotes the set of edges across multiple relation types. Specifically, $V = V_{drug} \cup V_{disease} \cup V_{protein}$, and $E = R_{DD} \cup R_{PP} \cup R_{DP} \cup R_{DS} \cup R_{PS} \cup R_{SS}$, where each relation set corresponds to a specific biological interaction type. Here, R_{DD} , R_{PP} , R_{DP} , R_{DS} , and R_{PS} correspond to known associations, namely drug-drug interactions, protein-protein interactions, drug-protein interactions, drug-disease associations, and disease-protein associations, respectively. The disease-disease associations R_{SS} were constructed by calculating the Jaccard similarity coefficient from both the drug-disease R_{DS} and protein-disease R_{PS} network (B. Wang, et al., 2023). In the drug-disease association network, the Jaccard similarity between S_i and S_j is defined as:

$$Similarity_{DS}(S_i, S_j) = \frac{|M_d(S_i) \cap M_d(S_j)|}{|M_d(S_i) \cup M_d(S_j)|} \quad (1)$$

Where $M_d(S_i)$ denotes the set of drugs associated with S_i . And S_i represents the i th disease. Similarly, in the protein-disease network, the similarity is calculated as:

$$Similarity_{PS}(S_i, S_j) = \frac{|M_p(S_i) \cap M_p(S_j)|}{|M_p(S_i) \cup M_p(S_j)|} \quad (2)$$

Where $M_p(S_i)$ denotes the set of proteins associated with S_i .

Based on these two similarity measures, the disease-disease association R_{SS} is established using a dual-threshold criterion:

$$R_{SS} = \begin{cases} 1, & \text{if } Similarity_{DS} > \theta \text{ and } Similarity_{PS} > \theta \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Where $\theta=0.7$ was selected based on empirical evaluation across a range of thresholds (from 0.3 to 0.9), with optimal performance observed at this value (see Table S1 of Appendix S1 for detailed results).

2.5. Learning node embeddings for biological entities by RGCN

In a heterogeneous biological network, different edge types convey distinct biological information, such as binding affinities in DTIs, therapeutic effects in DDAs, and functional or structural relationships in PPIs. To effectively learn node embeddings in such multi-relational graphs, we employed a 2-layer Relational Graph Convolutional Network (RGCN) (Schlichtkrull, et al., 2018). This architecture models various relation types through relation-specific transformation matrices, enabling the aggregation of both biological semantics and topological structures from heterogeneous neighborhoods (Tian, Zhang, Rang,

Yang, & Zhan, 2020). Specifically, each node updates its representation by computing a normalized sum of the embeddings of its neighbors, where the transformation applied to each neighbor depends on the relation type connecting them.

To improve parameter efficiency and mitigate overfitting, we incorporated basis decomposition into the RGCN framework. In this approach, each relation-specific weight matrix is expressed as a linear combination of a small set of shared basis matrices, thereby significantly reducing the total number of trainable parameters. The message-passing mechanism of RGCN is defined as:

$$h_i^{(l+1)} = \sigma \left(\sum_{r \in R} \sum_{j \in N_r^i} \frac{1}{|N_r^i|} W_r^{(l)} h_j^{(l)} + W_0^{(l)} h_i^{(l)} \right) \quad (4)$$

Where $h_i^{(l)}$ denotes the embedding of node i at layer l , R is the set of relation types, N_r^i represents the set of neighbors of node i under relation $r \in R$, $W_r^{(l)}$ and $W_0^{(l)}$ are learnable weight matrices, and σ is the activation function (we use LeakyReLU in our implementation) (J. Xu, Li, Du, Zhang, & Liu, 2020). The relation-specific transformation matrices $W_r^{(l)}$ are computed using basis decomposition:

$$W_r^{(l)} = \sum_{b=1}^B a_{rb}^{(l)} V_b^{(l)} \quad (5)$$

Where, B is the number of basis matrices, $V_b^{(l)}$ are the shared basis matrices, and $a_{rb}^{(l)}$ are the learnable coefficients for relation r and basis matrix b at layer l . The final node embedding for drugs, proteins, and disease are obtained after passing through the two layers of the RGCN encoder.

To predict potential associations between entities, we used the DistMult model (Yang, 2014) as the scoring function, which evaluated the plausibility of given triple (d, r', t) based on the learned embeddings:

$$f(d, r', t) = h_d^T R_{r'} h_t \quad (6)$$

Where $h_d, h_t \in \mathbb{R}^k$ denote the embedding vectors of the head entity (e.g., drug) and tail entity (e.g., protein or disease), respectively, and $R_{r'} \in \mathbb{R}^{k \times k}$ is a diagonal matrix associated with relation r' . DistMult is chosen for its simplicity, expressiveness in modeling symmetric relations common in biomedical networks, and compatibility with multi-relational graph learning. The resulting scores serve as logits in the subsequent weighted cross-entropy loss with negative sampling, enabling end-to-end training. Additionally, the learned node embeddings, which encode both biological semantics and network topology are reused as enriched node features for downstream tasks, including network refinement and link prediction.

2.6. Predicting DTIs and DDAs using Gelato

The original heterogeneous network, constructed from known public databases and manual curation, may be incomplete and contain missing or uncertain relationships, which can degrade link prediction performance (Zhaoyang Huang, Xiao, Ao, Guan, & Yu, 2025). To address this limitation, we adopted Gelato (Zexi Huang, et al., 2023), a graph enhancement framework that integrates node attribute information into the topological structure to improve link prediction accuracy. Gelato leverages the autocovariance heuristic to capture both local and global structural patterns in the network, effectively identifying latent relationships between node pairs, making it well-suited for link prediction tasks. The overall framework consists of two stages: graph enhancement and link prediction.

Given that our goal is to predict DDAs and DTIs, we first extract a subgraph of the original heterogeneous network that contains these interactions. The subgraph is denoted as $G_0 = \{V_0, E_0\}$ from the original heterogeneous network, where $V_0 = V$ is the full node set, and $E_0 = R_{DP} \cup R_{DS}$ contains only DTI and DDA edges. Following the guilt-by-association principle, we assume that nodes with similar embeddings

are more likely to form biologically meaningful associations. Accordingly, we enrich the graph by adding new edges based on the cosine similarity of node embeddings learned by RGCN. We define the enhanced edge set $\tilde{E}_0$ as:

$$\tilde{E}_0 = E_0 + \{(p, q) | \text{sim}(h_p, h_q) > \varepsilon_\eta, R_{pq} \in R_{DP}, R_{DS}\} \quad (7)$$

Where $\text{sim}(h_p, h_q)$ is the cosine similarity between embeddings of nodes p and q , and ε_η is a threshold chosen such that the number of added edges equals η times the original edge count, with $\eta \in \{0, 0.25, 0.5, 0.75, 1\}$.

Subsequently, we employ a Multi-Layer Perceptron (MLP) to assign weights to the edges in the enhanced graph $\tilde{G}_0 = \{V_0, \tilde{E}_0\}$. The MLP learns complex nonlinear relationships by combining node embedding features:

$$w_{pq} = \text{MLP}(h_p + h_q; |h_p - h_q|; \theta) \quad (8)$$

Where $[\cdot; \cdot]$ denotes vector concatenation, $h_p + h_q$ captures shared features, and $|h_p - h_q|$ (element-wise absolute difference) encodes dissimilarity, and θ represents the trainable parameters. Since DTIs and DDAs are undirected, we ensure symmetry by computing $w_{pq} = w_{qp}$.

The final adjacency matrix $\tilde{A}$ is a weighted combination of the original binary adjacency matrix A , the MLP weights W , and the cosine similarity matrix $\text{sim}(h_p, h_q)$:

$$\tilde{A}_{pq} = \alpha A_{pq} + (1 - \alpha)(\beta w_{pq} + (1 - \beta)\text{sim}(h_p, h_q)) \quad (9)$$

Where $\alpha, \beta \in [0, 1]$ are hyperparameters, and all continuous scores are normalized to $[0, 1]$ before fusion.

In the link prediction stage, the autocovariance algorithm is applied to capture the local and global topological relationships between node pairs within the enhanced graph. Autocovariance is a topological heuristic that calculates similarity by comparing the co-visiting probabilities of node pairs in truncated random walks with those in infinitely long random walks (Zexi Huang, Silva, & Singh, 2021). This makes it particularly effective at identifying latent edges by detecting statistically significant deviations from randomness. Given the enhanced graph $\tilde{G}_0$ with adjacency matrix $\tilde{A}$, the autocovariance similarity matrix $AR \in \mathbb{R}^{|n| \times |n|}$ is defined as:

$$AR = \frac{\tilde{D}}{\text{vol}(\tilde{G}_0)} (\tilde{D}^{-1} \tilde{A})^t - \frac{\tilde{d} \tilde{d}^T}{\text{vol}^2(\tilde{G}_0)} \quad (10)$$

Where $\tilde{D}$ is the diagonal degree matrix of $\tilde{A}$, $\tilde{d} = \text{diag}(\tilde{D})$ is the degree vector, $\text{vol}(\tilde{G}_0) = \sum_i \tilde{d}_i$ is the total volume of graph $\tilde{G}_0$, t is the number of steps in the truncated random walks. Each entry AR_{pq} quantifies the topological similarity between nodes p and q , with higher values indicating a higher likelihood of an unobserved interaction.

2.7. The composite loss function with uncertainty weighting

The entire training process of RGGE-DTD involves learning node embeddings using RGCN and predicting associations using Gelato. To jointly optimize these components in a multi-task setting, we design a composite loss function that combines both classification and ranking objectives, with automatic weighting based on task uncertainty.

First, for the RGCN-based prediction module, we treat DTI and DDA prediction as binary classification tasks. The output of the DistMult scoring function is passed through a sigmoid layer to produce a probability estimate of interaction. To address the severe class imbalance (i.e., far fewer known interactions than non-interactions), we employ a weighted cross-entropy loss for each task. For drug-target interactions (DTI), the loss is defined as:

$$\mathcal{L}_{\text{cross}}^{\text{DP}} = \frac{1}{|N_{\text{DP}}|} \sum_{(d, r', t, y) \in E_{\text{DP}}} (w^+ y \log \sigma(f(d, r', t)) + w^- (1 - y) \log(1 - \sigma(f(d, r', t)))) \quad (11)$$

Where E_{DP} is the set of DTI samples, $|N_{\text{DP}}|$ is the total number of positive and negative samples, $y \in \{0, 1\}$ is the ground truth label, $f(d, r', t)$ is the DistMult score, σ is the sigmoid function, and the class weights are set as $w^+ = \frac{N_{\text{DP}}^-}{N_{\text{DP}}}$ and $w^- = \frac{N_{\text{DP}}^+}{N_{\text{DP}}}$ to balance positive and negative contributions. A similar $\mathcal{L}_{\text{cross}}^{\text{DS}}$ is defined for drug-disease associations.

Second, for the Gelato module, we used the N-pair loss to improve the discriminative power of the learned similarity scores. This loss enhances ranking performance by contrasting each positive pair (p, q) with multiple negative pairs sampled from the same batch, providing a stronger supervisory signal under extreme class imbalance (Sohn, 2016). The N-pair loss for DTI is:

$$\mathcal{L}_{\text{N-pair}}^{\text{DP}} = - \sum_{(p, q) \in E_{\text{DP}}} \log \frac{\exp(AR_{pq})}{\exp(AR_{pq}) + \sum_{(u, v) \in N(p, q)} \exp(AR_{uv})} \quad (12)$$

Where AR_{pq} is the autocovariance similarity score between nodes p and q , $N(p, q)$ denotes the set of negative pairs in the mini-batch. The N-pair loss $\mathcal{L}_{\text{N-pair}}^{\text{DP}}$ is minimized when each positive edge (p, q) has a much higher similarity than its contrasted negative pairs: $AR_{pq} \gg AR_{uv}, \forall (u, v) \in N(p, q)$. A corresponding loss $\mathcal{L}_{\text{N-pair}}^{\text{DS}}$ is computed for DDA.

To combine these four losses- $\mathcal{L}_{\text{cross}}^{\text{DP}}$, $\mathcal{L}_{\text{cross}}^{\text{DS}}$, $\mathcal{L}_{\text{N-pair}}^{\text{DP}}$, and $\mathcal{L}_{\text{N-pair}}^{\text{DS}}$ - we adopt an uncertainty-aware multi-task loss formulation (Kendall, Gal, & Cipolla, 2018). This approach treats each subtask as an independent regression under Gaussian noise and learns task-dependent uncertainty parameters to automatically balance the contribution of each loss. The final composite loss is:

$$\mathcal{L}_{\text{composite}} = \frac{1}{2\beta_1^2} \mathcal{L}_{\text{cross}}^{\text{DP}} + \frac{1}{2\beta_2^2} \mathcal{L}_{\text{cross}}^{\text{DS}} + \frac{1}{2\beta_3^2} \mathcal{L}_{\text{N-pair}}^{\text{DP}} + \frac{1}{2\beta_4^2} \mathcal{L}_{\text{N-pair}}^{\text{DS}} + \log \beta_1 \beta_2 \beta_3 \beta_4 \quad (13)$$

Where, $\beta_i, i \in \{1, 2, \dots, 4\}$ represents the learnable relative weights of the corresponding loss. To prevent the cross-entropy loss from dominating the other tasks, we initialized the weights of β_1 and β_2 higher than the others. Furthermore, we applied an exponential mapping to the total loss to ensure that the loss function remains within the positive domain, providing valid values for variance. The pseudo-code of RGGE-DTD is listed in **Algorithm 1** below.

Algorithm 1: RGGE-DTD

Input: heterogeneous network G , subgraph $G_0 \in G$, drug structure Seq_{drug} , target sequence $\text{Seq}_{\text{protein}}$, disease association features S_{fea}

Output: predict score of AR_{DP} , AR_{DS} for DTIs and DDAs

1. Initialize RGGE-DTD model parameters δ
2. for $i=1$ to epochs **do**
3. $X_D = \text{CNN}(\text{Seq}_{\text{drug}})$, $X_P = \text{CNN}(\text{Seq}_{\text{protein}})$, $X_S = \text{Node2Vec}(S_{\text{fea}})$
4. $X = \text{concatenate}(X_D, X_P, X_S)$
5. $H = \text{RGCN}(X, G)$
6. Compute the distMult score $f(d, r', t)$ for DTIs and DDAs with Equation (6)
7. $AR = \text{Gelato}(H, G_0)$
8. Compute $\mathcal{L}_{\text{cross}}^{\text{DP}}$, $\mathcal{L}_{\text{cross}}^{\text{DS}}$ with Equation (11)
9. Compute $\mathcal{L}_{\text{N-pair}}^{\text{DP}}$, $\mathcal{L}_{\text{N-pair}}^{\text{DS}}$ with Equation (12)
10. Compute the joint loss $\mathcal{L}_{\text{composite}}$ with Equation (13)
11. Update RGGE-DTD model parameters δ using $\mathcal{L}_{\text{composite}}$
12. Extract predict score AR_{DP} , AR_{DS} from AR
13. **end**
14. **Function** $\text{RGCN}(X, G)$
15. for $k=0$ to l layers:
16. Compute H^k with Equation (4)
17. $H^k = \text{LayerNorm}(H^k)$
18. $H^k = \text{LeakyReLU}(H^k)$
19. return H
20. **Function** $\text{Gelato}(H, G_0)$

(continued on next page)

(continued)

Algorithm 1: RGGE-DTD

-
21. Compute the enhanced edge set $\tilde{E}_0$ with Equation (7)
 22. Compute the edge weights w_{pq} with Equation (8)
 23. Compute the enhanced adjacency matrix $\tilde{A}_{pq}$ with Equation (9)
 24. Compute the autocovariance similarity matrix AR with Equation (10)
 25. return AR
-

3. Results

3.1. Unbiased training and testing

To ensure a fair and realistic evaluation of model performance, we adopted a strict data splitting strategy that prevents information leakage from the validation and test sets into the training phase (Zexi Huang, et al., 2023). This setup simulates real-world scenarios where new drug–disease associations (DDAs) or drug–target interactions (DTIs) are discovered over time, and the model must generalize to previously unseen positive interactions.

Specifically, for both DDAs and DTIs, we perform 10-fold cross-validation on the set of known positive interactions E^+ . In each of 10 rounds, one fold (approximately 10% of E^+) is held out as the test set E_{test}^+ , while the remaining nine folds (approximately 90% of E^+) constitute the training pool. From this training pool, we randomly reserve 5% of the total E^+ (i.e., roughly half of one fold) as the validation set E_{valid}^+ , and the rest (approximately 85%) constitutes the training set E_{train}^+ . This procedure is repeated independently across all 10 folds, ensuring that every positive interaction serves exactly once as a test example and multiple times in training or validation across different rounds. Let $\mathcal{Z}$ denote the set of all possible drug–target (or drug–disease) pairs, and define the set of truly unobserved pairs as $E^- = \mathcal{Z} \setminus E^+$. During a given CV round, the model only observes E_{train}^+ as positive examples. All other pairs—namely E_{valid}^+ , E_{test}^+ , and E^- —are unseen as positives and thus treated as unlabeled candidates. For negative sampling in training, we draw from the union $E^- \cup E_{valid}^+ \cup E_{test}^+$; similarly, validation negatives are sampled from $E^- \cup E_{test}^+$, and test negatives solely from E^- . To illustrate, consider a true drug–disease association $(d_1, dis_1) \in E_{test}^+$. During training, this pair is not revealed as a positive; instead, it resides among the unlabeled candidates and may be sampled as a negative example. The model therefore treats (d_1, dis_1) as non-interacting—mirroring the real-world scenario where such an association has not yet been discovered. Only at test time is its true label disclosed, allowing us to assess whether the model assigns it a high rank among all candidate pairs. This protocol ensures that no held-out positive is ever observed during its corresponding training phase, thereby preventing information leakage and enabling a realistic evaluation of the model's ability to prioritize novel associations.

For the unbiased testing process, instead of relying solely on AUC (which can be misleading in highly imbalanced settings) (J. Huang & Ling, 2005), we adopted three rank-based metrics that are more informative for sparse and skewed datasets: precision@K (Lü & Zhou, 2011), hits@K (Bordes, Usunier, Garcia-Duran, Weston, & Yakhnenko, 2013), and AUPR (Saito & Rehmsmeier, 2015). These metrics evaluate how well the model ranks true positive interactions against all possible negative candidates, providing a robust assessment of retrieval performance.

Precision@K computes the proportion of true positives among the top K interactions. The formula is as follows:

$$\text{Precision@K} = \frac{TP_K}{K}$$

Where TP_K is the number of true positives within the top K predictions.

Hits@K indicates whether at least one true positive appears in the top K predictions. For a single query, it is defined as:

$$\text{Hits@K} = \begin{cases} 1 & \text{if } TP_K \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

The final Hits@K score is reported as the **average** over all test queries, reflecting the model's ability to retrieve relevant interactions.

AUPR computes the area under the precision-recall curve, integrating precision across all recall levels:

$$\text{AUPR} = \sum_{k=1}^n (R_k - R_{k-1}) * P_k$$

Where R_k and P_k are the recall and precision at the k-th threshold, and n is the number of different thresholds. AUPR is particularly informative in highly imbalanced settings, as it focuses on the performance of positive samples.

These metrics provide a comprehensive view of the model's predictive accuracy, especially in identifying rare but meaningful interactions. By combining rigorous data splitting with rank-based evaluation metrics, our protocol provides a reliable and unbiased assessment of the model's generalization capability in predicting novel DDAs and DTIs.

3.2. Hyperparameter Settings and Experimental Setup

In this study, we optimized the RGGE-DTD model by fine-tuning the hyperparameters of its constituent models. To construct the disease-disease network, we adjusted the similarity threshold θ between two networks from the set $\{0.3, 0.5, 0.7, 0.9\}$. We then employed Optuna (Akiba, Sano, Yanase, Ohta, & Koyama, 2019), a Bayesian optimization framework, to identify the best settings based on prior evaluations. The embedding dimensions of the datasets were set to $\{32, 64, 128\}$, and the depth of layers for both RGCN and MLP was varied from $\{1, 2, 3\}$. The hidden layer dimensions of the model and the output dimension of the final MLP layer were selected from $\{32, 64, 128\}$. In Gelato, the threshold η for adding edges was set to $\{0.0, 0.25, 0.5, 0.75, 1.0\}$, while the weighted coefficient α and β for enhanced networks were set to $\{0.0, 0.25, 0.5, 0.75\}$ and $\{0.25, 0.5, 0.75, 1.0\}$, respectively. The learning rate was selected from $\{0.0001, 0.0005, 0.001, 0.002, 0.005, 0.01\}$. The range selection of these parameters follows the guidelines in a previous study (Zexi Huang, et al., 2023). The process of adjusting hyperparameters and the best combinations employed in this study are listed in Tables S2, S3 of Appendix S1.

We recorded the actual running time and runtime complexity of the RGCN and Gelato modules on our dataset, and the results demonstrate the practical efficiency of the algorithm in handling large-scale heterogeneous network data (See Appendix S2). All experiments were conducted on a desktop workstation running Microsoft Windows 11 Pro (Version 10.0.26100) equipped with an Intel® Core™ i7-14700KF processor (3.4 GHz, 20 cores, 28 threads) and 64 GB of RAM, and an NVIDIA GeForce RTX 4080 SUPER GPU with 16 GB of VRAM.

3.3. Comparison experiments

To evaluate the predictive performance of our proposed RGGE-DTD model in both DDAs and DTIs prediction tasks, we conducted comprehensive comparisons against ten state-of-the-art baseline methods. These including five DDA-specific models: deepDR (Zeng, et al., 2019), LAGCN (Yu, et al., 2021), DDAGDL (Zhao, Su, et al., 2022), HINGRL (Zhao, Hu, et al., 2022), DTD-MTL (B. Wang, et al., 2023) and six DTI-specific models: DTINet (Y. Luo, et al., 2017), NeoDTI (Wan, et al., 2019), DTI-CNN (Peng, et al., 2020), AMGDTI (Su, et al., 2024), DTD-MTL (B. Wang, et al., 2023), CSCL-DTI (Lin, et al., 2024). Notably, DTD-MTL is a multitask model capable of predicting both DDAs and DTIs, and thus was evaluated in both scenarios. The baselines were selected based on two criteria: (1) architectural relevance and methodological diversity, including recent graph neural networks (e.g., LAGCN, DDAGDL, HINGRL), meta-graph learning (AMGDTI), multi-task

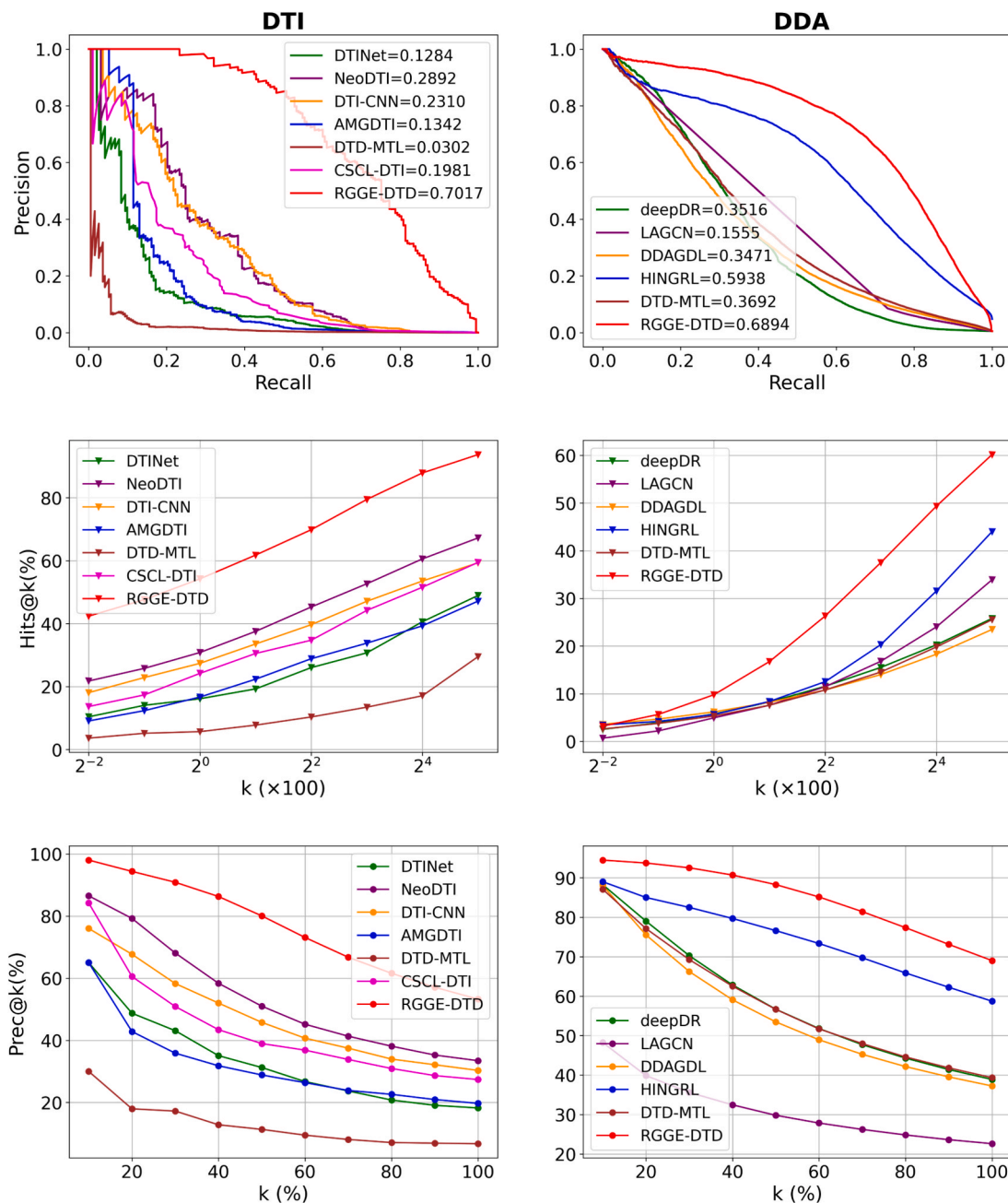


Fig. 2. Performance comparison of six methods in unbiased metrics (PR curves, hits@k, and prec@k) for DTIs and DDAs predictions. The PR curve illustrates the relationship between Precision and Recall across different thresholds. The legend uses the AUPR value, which represents the area under the precision-recall curve. The Hits@k and Prec@k show the results of hits@k (with k as the rank) and prec@k (with k as the ratio of test edges) based on varying values of k.

frameworks (DTD-MTL), and dual-view contrastive approaches (CSCL-DTI); and (2) widespread adoption and benchmarking value in prior studies (e.g., DTINet, deepDR). All models were evaluated on the same benchmark dataset (Y. Luo, et al., 2017) using 10-fold cross-validation, with identical train/validation/test partitions and negative sampling following the unbiased protocol described in Section 3.1. Default hyperparameters from the original studies were used to ensure a fair and reproducible comparison.

Comparison in unbiased metrics: Fig. 2 represents the performance comparison in terms of unbiased metrics—Precision@K, Hits@K, and AUPR—which are particularly informative for highly imbalanced datasets. RGGE-DTD achieves superior performance in both tasks, with AUPR values of 0.7017 for DTI prediction and 0.6894 for DDA prediction.

In DTI prediction, NeoDTI ranked second, as it considers the heterogeneity of distinct edge types and aggregates neighborhood information for each edge type, similar to our approach. However, NeoDTI relies on the observed network, which may contain noisy or missing edges that degrade its performance, resulting in an AUPR approximately 0.42 lower than ours. DTI-CNN, which learns representations based on the similarities of different individual networks, fails to capture information across heterogeneous networks, leading to an AUPR approximately 0.06 lower than NeoDTI. CSCL-DTI also employs contrastive learning to enhance cross-view representation learning; however, it utilizes only structural features of drugs and proteins while ignoring heterogeneous biological networks, leading to a performance drop of nearly 0.04 in AUPR compared to DTI-CNN. By comparison, AMGDTI, DTINet, and DTD-MTL show limited performance (AUPR < 0.15),

potentially due to reliance on pre-defined meta-paths (AMGDTI), oversimplified diffusion processes (DTINet), or ineffective task coordination in multi-task learning (DTD-MTL). For DDA prediction, HINGRL achieves the best performance among baselines (AUPR=0.5938) by combining biological and topological features, though it treats them independently and ignores their intrinsic relationship. DeepDR (AUPR=0.3516) leverages deep autoencoders across multiple networks but does not explicitly incorporate biological node attributes. DTD-MTL and DDAGDL, both based on relational GNNs, achieve AUPR of 0.3692 and 0.3471, respectively, but rely heavily on existing networks. Notably, DTD-MTL's performance in DDA prediction is significantly better than in DTI prediction, likely due to the challenges in hyperparameter tuning for coordinating multiple tasks. LAGCN, which uses GAT on only three subnetworks, exhibits the lowest performance (AUPR=0.1555), suggesting that limited contextual information constrains its predictive power. Consistent trends are observed in Precision@K and Hits@K, confirming the robustness of RGGE-DTD's advantage.

The superior performance of RGGE-DTD can be attributed to several key innovations. First, it enhances the input heterogeneous information network (HIN) through representation-guided link inference, improving structural completeness. Second, it integrates rich biological features (e.g., drug fingerprints, protein sequences) with topological context in a unified framework. Third, it employs autocovariance to capture high-order topological dependencies. Finally, it employs uncertainty-based weighting to automatically balance the multi-task losses, thereby eliminating the need for manual tuning. In contrast, many existing methods either ignore biological node attributes (e.g., deepDR, LAGCN), fail to fuse topological and biological information (e.g., DTI-CNN, HINGRL, CSCL-DTI), or struggle with multi-task coordination (e.g., DTD-MTL). In summary, RGGE-DTD outperforms state-of-the-art methods by jointly addressing challenges of network incompleteness, feature integration, and multi-task optimization—demonstrating its effectiveness in predicting both DDAs and DTIs.

Comparison in biased metrics: since most existing methods employ biased evaluation protocols—such as balanced training sets with equal numbers of positive and negative samples—we conducted additional experiments under the identical conditions to ensure fair comparison. Specifically, we constructed a balanced dataset by randomly sampling non-interacting drug–target pairs to match the number of known interactions. The 10-fold cross-validation was then performed on this dataset according to the protocol in Section 3.1. Under this balanced setting, we computed the ROC curves and reported AUC scores for all models (Fig. 3). Additionally, we applied a paired t-test to assess whether RGGE-DTD's gains over baseline methods are statistically significant (Table 2).

As expected, all models showed inflated AUC values under this

Table 2

Paired t-test(p-values) for RGGE-DTD versus baseline methods in DTI and DDA prediction tasks. All evaluations follow an unbiased 10-fold cross-validation protocol.

DTIs	p-value	DDAs	p-value
DTINet	3.12×10^{-5}	deepDR	2.09×10^{-7}
NeoDTI	2.56×10^{-4}	DTD-MTL	1.08×10^{-8}
AMGDTI	8.86×10^{-3}	HINGRL	6.76×10^{-7}
DTI-CNN	5.59×10^{-4}	LAGCN	6.30×10^{-6}
DTD-MTL	1.74×10^{-4}	DDAGDL	6.45×10^{-9}
CSCL-DTI	6.13×10^{-5}	—	—

biased evaluation framework. Our model achieved the highest performance, with AUCs of 0.9969 for DTI and 0.9955 for DDA prediction. Notably, several methods—including DTINet, AMGDTI, CSCL-DTI, and DTD-MTL for DTIs; and deepDR, LAGCN for DDAs—performed poorly under unbiased (imbalanced) testing but showed strong results under biased testing. This discrepancy suggests that these models are highly sensitive to class distribution. The underlying cause may lie in commonly adopted design choices, such as learning representations based primarily on node features, relying on network diffusion processes, applying aggressive dimensionality reduction, or employing multi-task learning frameworks that fail to generalize well under realistic data imbalance. In contrast, our model maintains superior performance even when the training data is artificially balanced, indicating robustness to evaluation protocol variations.

Nevertheless, high performance under artificially balanced settings does not guarantee effectiveness in real-world drug discovery, where known interactions constitute only a tiny fraction of all possible drug–target pairs, and no prior knowledge enables balanced sampling. To assess practical utility under more realistic and challenging conditions, we further evaluated our model using the new drug and singleton drugs prediction protocol proposed by DTINet (Y. Luo, et al., 2017). Specifically, for the new drug setting, we identified 89 drugs with newly annotated mechanisms of action in DrugBank v4.5, which were not present in v3.0 used for network construction. A test set of 521 DTIs involving these drugs was held out, and models were trained on all other interactions. This setup evaluates the model's ability to predict targets for previously unseen drugs. Following DTINet, we used Recall@k as the evaluation metric, defined as the fraction of true interacting targets ranked within the top-k predictions for each drug. As shown in Fig. S2(a) of Appendix S3, RGGE-DTD achieved significantly higher recall across all values of k compared to baseline methods, indicating its superior capability in prioritizing true targets for novel drugs. In the singleton drug setting, we identified 181 drugs each associated with exactly one known DTI in the dataset. During training, the single known interaction

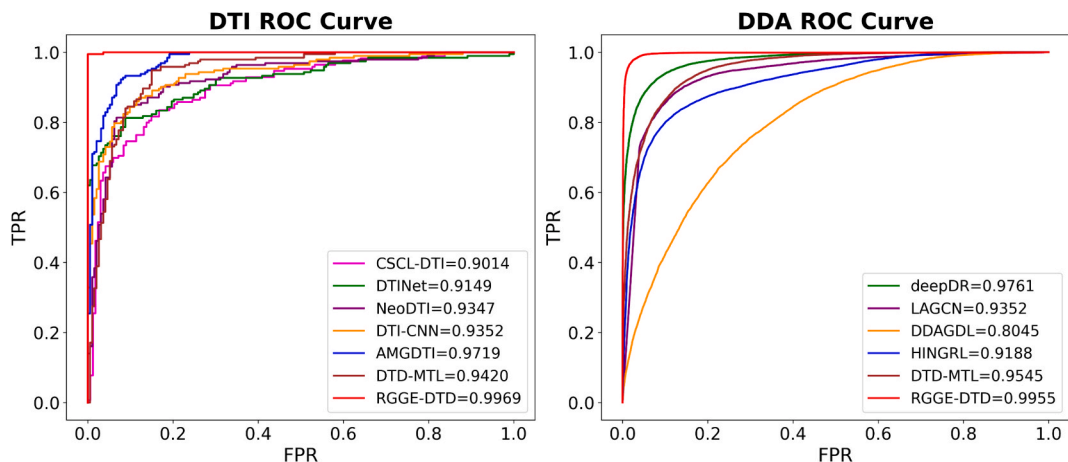


Fig. 3. Performance comparison of six methods in biased metrics (ROC curve) for DTIs and DDAs predictions.

for each singleton drug was excluded and reserved as the test label, simulating a cold-start scenario for poorly characterized compounds. As illustrated in Fig. S2(b) of Appendix S3, RGGE-DTD retrieved approximately 85% of true DTIs within the top 150 predictions for these 181 singleton drugs, substantially outperforming the second-best method (AMGDTI), which achieved less than 15% recall at the same rank.

These results demonstrate that while many existing methods benefit from artificially balanced data, their performance degrades significantly under realistic, imbalanced, and cold-start conditions. In contrast, RGGE-DTD excels in these challenging scenarios, effectively leveraging heterogeneous biological and topological information to generalize to both novel and sparsely annotated drugs. The paired-t test further confirmed that RGGE-DTD achieves highly significant improvements over all baseline methods in both DTI and DDA prediction tasks (all $p < 0.001$; Table 2). Therefore, although biased evaluation protocols may yield high AUC scores, they can mask poor generalization and overestimate model performance. Our findings suggest that unbiased, imbalanced evaluation settings are more appropriate for assessing the true predictive power of computational models in drug discovery.

3.4. Impact of loss function and training strategy

In this section, we systematically evaluate the efficacy of the proposed composite loss function and unbiased training strategy for predicting the average association strength of DTIs and DDAs. To this end, we conducted comprehensive experiments using two distinct composite loss formulations: (1) a cross-entropy (CE) loss applied uniformly across both the RGCN and the Gelato model, and (2) a hybrid joint loss combining cross-entropy for RGCN training with an N-pair (NP) loss for Gelato model training, similar to our proposed approach. These experiments were meticulously designed to compare the performance under biased and unbiased conditions.

Fig. 4 illustrates the training and validation losses, as well as the precision at 100% (prec@100%), for the RGGE-DTD model under both conditions. The final test AUPR scores are reported in the subplot titles. Under biased conditions, the validation precision fluctuates more with the CE+CE configuration than with CE+NP, indicating that the purely classification-based loss is less stable when generalizing to unbiased test conditions. In contrast, the integration of NP loss—which explicitly encourages discriminative embedding learning by pulling positive pairs closer and pushing negative ones apart—complements the CE loss by capturing inter-sample similarity structures. This leads to improved consistency across datasets and a 3% higher AUPR compared to CE+CE.

Under unbiased training, both configurations exhibit lower precision during validation due to the increased number of negative samples and the resulting rise in false positives. Nevertheless, the CE+NP configuration achieves approximately 12% higher AUPR than CE+CE, demonstrating its superior robustness to class imbalance. This improvement stems from the NP loss's ability to jointly optimize the relative distances among positive and negative samples, thereby enhancing the model's capacity to distinguish true associations from the vast pool of non-interactions. In contrast, the CE loss treats each sample independently in terms of classification probability, without explicitly modeling pairwise relationships. Notably, for both loss configurations, the AUPR under unbiased training significantly exceeds that under biased training. This counterintuitive result highlights the limitation of biased evaluation: artificially balancing classes can distort the optimization objective and degrade generalization to real-world imbalanced scenarios. In contrast, AUPR—a metric that emphasizes performance at high recall—is particularly sensitive to class imbalance and provides a more realistic assessment of model utility in drug discovery.

Collectively, unbiased training accelerates model learning and enhances information capture, while NP+CE loss outperforms CE+CE loss by effectively discriminating negative samples and identifying potential false negatives in unseen data. These findings underscore the importance of aligning training protocols with real-world application scenarios.

3.5. Contribution of individual subnetwork

To evaluate the contribution of each biological knowledge source to the overall performance of RGGE-DTD, we conducted a systematic ablation study by sequentially removing one sub-network from the heterogeneous graph while keeping all other components unchanged. All models were trained under the unbiased protocol, and performance

Table 3

The impact of different subnetworks on model training

Networks	DTIs	DDAs
Without drug-drug	0.6604	0.6863
Without drug-protein	0.5228	0.5726
Without drug-disease	0.5921	0.4856
Without protein-protein	0.6674	0.6856
Without protein-disease	0.6710	0.6811
Without disease-disease	0.6665	0.6744
Heterogeneous network	0.7017	0.6894

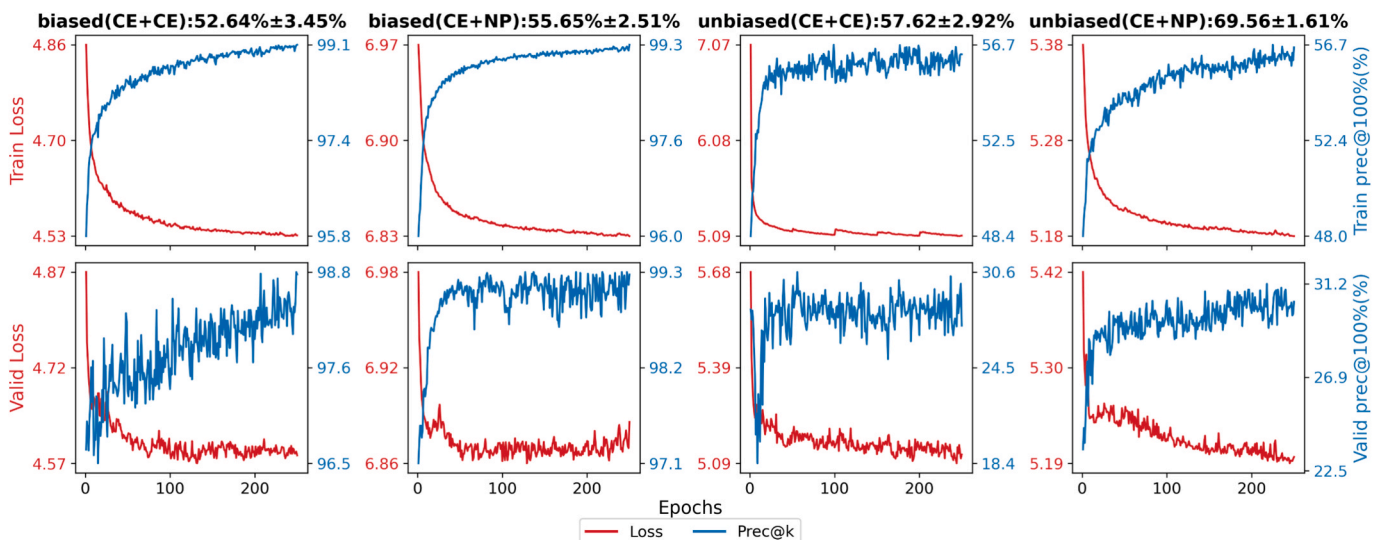


Fig. 4. Training of RGGE-DTD based on different losses and training settings, with test AUPR(mean±std) shown in the titles.

was assessed using AUPR on the test set. Results are summarized in Table 3.

The overall performance of the heterogeneous network is superior in predicting both DTIs and DDAs. As expected, the removal of directly task-relevant networks led to the most significant performance degradation. For DTI prediction, excluding the drug–protein interaction(DTI) network resulted in an AUPR drop of 0.18, while removal of the drug–disease association(DDA) network caused a decrease of 0.11. Similarly, for DDA prediction, eliminating the DDA network reduced AUPR by 0.20, and removing the DTI network led to a decline of 0.13. These substantial drops confirm that the DTI and DDA networks provide essential, high-value signals for their respective prediction tasks—reflecting known pharmacological mechanisms and therapeutic indications. Notably, cross-task information transfer is also evident: the DTI network significantly contributes to DDA prediction, and vice versa. This supports the effectiveness of our joint learning framework in leveraging shared representations across related tasks. In contrast, the removal of other sub-networks—such as protein–protein interactions, drug–drug similarities, disease–disease semantic similarities, and gene–disease associations—resulted in only minor performance changes ($\Delta\text{AUPR} < 0.04$), indicating their supplementary rather than critical role in this specific context. Despite the loss of certain biological modalities, RGGE-DTD maintains robust predictive performance, with no single non-core network removal causing a catastrophic failure. This demonstrates the model’s resilience to partial data incompleteness—a critical advantage in real-world applications where biological annotations often incomplete or noisy.

3.6. Ablation experiment

To investigate the impact of model components and input node features on the performance of RGGE-DTD, we conducted ablation experiments by removing distinct model components and replacing the input node features. Table 4 shows the AUPR values for various combinations using an unbiased dataset split and 10-fold cross-validation. The input data variants include two types: (1) biological feature used by RGGE-DTD, which utilized the original domain-specific features of nodes, such as molecular fingerprints for drugs, amino acid sequences for proteins, and semantic information for diseases; and (2) similarity feature, which employed the similarity calculation derived by computing pairwise similarities from the original features, followed by principal component analysis (PCA) to obtain low-dimensional representations. The model components ablation experiments involved removing one of the core components or prediction tasks, including ablating a single task (i.e., retaining only either DTI or DDA predictions), ablating the RGCN module, and ablating the Gelato module.

As shown in Table 4, the full model achieves the highest AUPR across both tasks(DTI: 0.7017; DDA: 0.6894). For direct comparison, the single-task baselines—trained exclusively on DTI or DDA labels—achieve AUPR scores of 0.5726 and 0.6356, respectively. This demonstrates that joint training yields absolute improvements of +0.1291 (+22.5%) for DTI and +0.0538 (+8.5%) for DDA, confirming that multi-task coupling provides genuine, quantifiable benefits rather than marginal gains.

Table 4

AUPR values after removal or replacement model and data components. “DTIs/DDAs” rows show single-task baselines (i.e., RGGE-DTD with one task disabled); All other rows represent ablations of model components, not task configurations.

MetricsAblation Target	Biological feature		Similarity feature	
	DTIs	DDAs	DTIs	DDAs
DTIs/DDAs	0.5726	0.6356	0.5379	0.5623
RGCN	0.5552	0.5243	0.4899	0.5313
Gelato	0.1063	0.3619	0.0931	0.1356
RGGE-DTD	0.7017	0.6894	0.5998	0.5686

Moreover, the performance degradation is asymmetric: removing the DDA task reduces DTI AUPR by 0.1291, whereas removing the DTI task reduces DDA AUPR by only 0.0538. This indicates that drug–disease association modeling contributes more substantially to drug–target prediction than the reverse, likely because known disease indications enrich drug representations with therapeutic context that informs potential mechanisms of action. Notably, ablating the Gelato module results in the most significant drop—reducing DTI AUPR to 0.1063 and DDA AUPR to 0.3619—indicating its crucial role in refining edge predictions through higher-order structural patterns such as shared neighborhoods and transitive connectivity. Ablating the RGCN module also reduces performance, demonstrating its ability to capture meaningful information from the weighted heterogeneous network topology and biological node features. At the feature level, models using raw biological features consistently outperform those using similarity-derived PCA representations (average $\Delta\text{AUPR} = +0.11$), indicating that dimensionality reduction and similarity transformation may discard fine-grained discriminative information critical for accurate link prediction.

In summary, our ablation study demonstrates that: (i) both RGCN and Gelato are essential for optimal performance, with Gelato playing a particularly vital role in structural refinement; (ii) multi-task learning enables beneficial knowledge transfer between DTI and DDA tasks; and (iii) native biological features preserve more predictive signal than transformed similarity embeddings. These findings underscore the importance of integrating domain-informed inputs with architecture-aware modeling in heterogeneous biomedical graphs.

3.7. Case study

To assess RGGE-DTD’s potential for discovering novel interactions, we focused on high-confidence predictions that are structurally and semantically dissimilar to known associations. Specifically, we excluded any candidate pair with drug chemical similarity > 0.6 and either (i) protein sequence similarity > 0.4 (for DTIs) or (ii) disease Jaccard similarity > 0.7 (for DDAs). From the full set of candidate drug–protein and drug–disease pairs, we identified the top 80 highest-scoring predictions for each task. As shown in Fig. 5, the majority of these top predictions correspond to known interactions (black lines), with only 2 drug–protein and 2 drug–disease pairs (red lines) not present in current databases. This high concentration of known interactions among top-ranked predictions confirms that RGGE-DTD effectively prioritizes biologically plausible associations, even when evaluated under an unbiased split.

To further evaluate RGGE-DTD’s ability to propose biologically plausible candidate associations not cataloged in existing knowledge bases, we analyzed the top 10 highest-scoring predicted drug–protein and drug–disease interactions absent from established databases. As summarized in Table 5, all 10 predicted drug–protein interactions are supported by independent experimental or clinical evidence. For instance, pancuronium—a non-depolarizing neuromuscular blocker—has been reported to exhibit affinity for muscarinic acetylcholine receptors (e.g., CHRM1) in cardiac tissues, potentially contributing to its cardiovascular side effects (Leung & Mitchelson, 1982). Meperidine exerts its analgesic and antishivering effects primarily through the μ -opioid receptor (OPRM1), modulating hypothalamic thermoregulation and neurotransmitter release (Kurz, et al., 1993). Mivacurium, another short-acting neuromuscular blocker, competitively inhibits nicotinic acetylcholine receptors (e.g., CHRNA1/CHRNA1) at the neuromuscular junction, thereby preventing muscle depolarization (Cook & Simons, 2023). Misoprostol, a synthetic prostaglandin E1 analog, activates the prostaglandin E receptor1 (PTGER1) to suppress gastric acid secretion and enhance mucosal defense (Oliva, et al., 2006). Additionally, terbutaline has been shown to enhance cardiac contractility via the β 1-adrenergic receptor (ADRB1), supporting its use in certain cardiovascular conditions (Behr, Hoffmann, Ottolina, & Klotz, 2006).

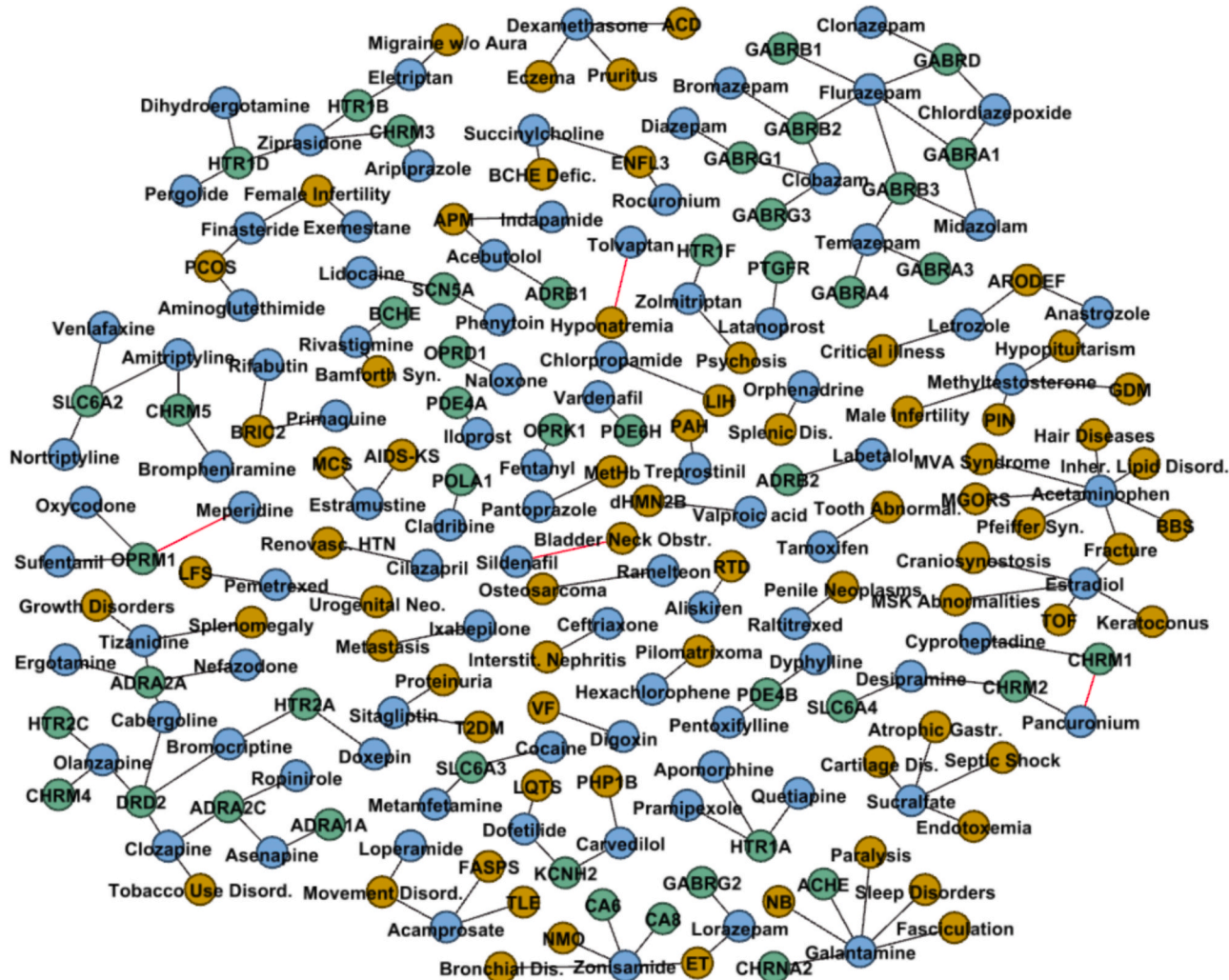


Fig. 5. The top 80 drug-protein interactions (DTIs) and drug-disease associations (DDAs) predicted by RGGE-DTD. Blue, brown, and green nodes represent drugs, targets, and diseases, respectively. Black and red lines indicate known and predicted interactions/associations, respectively.

Table 5
10 novel drug-protein interaction among the list of top 10 significant predictions derived by RGGE-DTD that can be supported by previous studies in the literature.

Drug name	Target name	Supporting references
Pancuronium	CHRM1	(Leung & Mitchelson, 1982)
Meperidine	OPRM1	(Kurz, et al., 1993)
Mivacurium	CHRM1	(Cook & Simons, 2023)
Pentazocine	OPRD1	(Shu, et al., 2011)
Rizatriptan	HTR1A	(Y.-C. Xu, et al., 1999)
Acetaminophen	ESR1	(Moreira et al., 2023)
Misoprostol	PTGER1	(Oliva, et al., 2006)
Dopamine	ADRA2A	(Zhang, Klimek, Farley, Zhu, & Ordway, 1999)
Terbutaline	ADRB1	(Behr, et al., 2006)
Ergotamine	HTR1A	(Bigal & Tepper, 2003)

Similarly, among the top 10 predicted drug-disease associations (Table 6), 9 are corroborated by existing literature. Tolvaptan, a selective vasopressin V2-receptor antagonist, effectively corrects hyponatremia by promoting aquaresis, as validated in multiple clinical trials (Tzoulis, Kaltsas, Baldeweg, Bouloux, & Grossman, 2023). Sildenafil alleviates bladder outlet obstruction by relaxing smooth muscle through the nitric oxide/cGMP pathway (Gur, Yurdaarmagan, Bayatli, & Sikka, 2016). Diazoxide, a KATP channel opener, has been linked to ventricular arrhythmias in type 2 diabetes via sarcolemmal KATP channel activation

Table 6
9 novel drug-disease association among the list of top 10 significant predictions derived by RGGE-DTD that can be supported by previous studies in the literature.

Drug name	Disease name	Supporting references
Tolvaptan	hyponatremia	(Fang, et al., 2016)
Sildenafil	urinary bladder neck obstruction	(Gur, et al., 2016)
Conivaptan	adrenal hyperplasia, congenital	—
Diazoxide	tachycardia, ventricular	(Xie, et al., 2015)
Acetaminophen	endocrine system diseases	(Lu, et al., 2024)
Acetaminophen	silver-russell syndrome	(Bartholdi, et al., 2009)
Galantamine	spina bifida cystica	(Rogers, Tvrdik, Capecechi, & Gahring, 2012)
Topiramate	glaucoma, open-angle	(Aminlari, et al., 2008)
Acetaminophen	vitamin d deficiency	(RC, G., 1998)
Galantamine	epilepsy, generalized	(Helbig, et al., 2009)

(Xie, Hu, Motloch, Karam, & Akar, 2015). Acetaminophen may disrupt ovarian function by modulating the SIRT-1/Nrf2/NF-κB signaling axis, potentially contributing to endocrine disorders (Lu, et al., 2024). Topiramate has been associated with acute angle-closure glaucoma due to ciliochoroidal effusion, as documented in multiple case reports (Aminlari, East, Wei, & Quillen, 2008).

Collectively, the strong concordance between RGGE-DTD's high-confidence predictions and established pharmacological knowledge underscores the model's ability to prioritize biologically meaningful interactions, even when trained on incomplete networks.

4. Discussion and conclusion

Predicting drug–target interactions (DTIs) and drug–disease associations (DDAs) is pivotal in drug repositioning, enabling the discovery of novel therapeutic uses for existing drugs. Traditional computational approaches to DTI and DDA prediction are often developed in isolation and constrained by limited prior knowledge. Recent advances in geometric deep learning have led to powerful frameworks such as HGCL-DR (G. Wang, Chen, et al., 2025), which employs heterogeneous graph contrastive learning with graph diffusion to predict drug–disease associations by integrating global and local feature representations, and MMPD-DTA (G. Wang, Zhang, Shao, Sun, & Cao, 2025) which integrates multi-modal deep learning with pocket–drug graphs to model atomic interactions for drug–target binding affinity prediction. Among these, network-based methods have emerged as particularly promising, as they leverage heterogeneous biological networks to uncover latent relationships and reveal therapeutic opportunities that may remain hidden in conventional, siloed analyses. Nevertheless, the inherent noise and complexity of heterogeneous networks, coupled with severe class imbalance between positive and negative associations, pose significant challenges for accurate prediction.

To address these challenges, we propose RGGE-DTD, a novel framework that jointly predicts DTIs and DDAs by deeply integrating enhanced heterogeneous networks with rich node features. RGGE-DTD employs a relational graph convolutional network (RGCN) to capture entity- and relation-specific semantics from the heterogeneous graph. To mitigate noise and uncertainty in the input networks, it incorporates the Gelato method to refine node representations using structural and semantic context. High-order topological patterns are learned to construct a pairwise scoring matrix, while a hybrid composite objective function enables effective training under unbiased evaluation settings, explicitly addressing class imbalance.

Extensive experiments demonstrate the superior performance and robustness of RGGE-DTD. It consistently outperforms ten state-of-the-art baselines in both DTI and DDA prediction tasks. Notably, performance discrepancies between biased and unbiased evaluation protocols reveal a pervasive optimism in many existing methods—a bias often overlooked but critically important for real-world applicability. Training dynamics and loss curves further confirm the effectiveness of the joint objective under unbiased settings, as reflected in higher AUPR scores and smoother convergence behavior. Analyses of individual network contributions highlight the complementary roles of the DTI and DDA subnetworks and underscore the model's resilience even when one network is removed. Ablation studies further validate the design choices: the Gelato module contributes most significantly to performance, followed by the RGCN encoder and the multi-task learning architecture. Moreover, biological features derived directly from original biomedical data (e.g., protein sequences, drug SMILES) consistently outperform similarity-based features, underscoring their richer information content. Finally, case studies highlight high-confidence predictions—such as the association between pancuronium and CHR11, and between tolcapten and hyponatremia—that align with established pharmacological knowledge, reinforcing the model's predictive validity.

Despite its strong performance, RGGE-DTD has several limitations. First, it is currently restricted to protein-coding targets and small-molecule drugs, excluding emerging therapeutic modalities such as non-coding RNAs (e.g., microRNAs), which are increasingly recognized as viable targets for pharmacological modulation. Second, predictions are context-agnostic; the model does not incorporate tissue-specific or disease-state multi-omics data (e.g., transcriptomic or proteomic profiles), which may limit its precision in biologically specific scenarios.

Third, molecular representations are limited to one-dimensional (1D) sequences and two-dimensional (2D) molecular graphs, omitting three-dimensional (3D) structural information that is crucial for modeling spatial binding interactions (Abbasi, et al., 2025). Finally, while RGGE-DTD excels at prioritizing candidate interactions, experimental validation remains essential to confirm biological activity and assess therapeutic potential.

These limitations point to promising avenues for future work: (1) extending the framework to non-coding RNA targets, (2) integrating context-aware multi-omics and 3D structural data to enable tissue- and structure-informed predictions, and (3) collaborating with wet-lab researchers to experimentally validate top-ranked candidates—thereby bridging computational discovery with translational impact.

Funding

This work was supported by the National Natural Science Foundation of China (Grant No.62202415), the Yunnan Fundamental Research Projects (Grant No. 202401AT070279; 202401AT070280; 202501AT070455), the Talent Introduction Project of Yunnan University of Finance and Economics (Grant No. 2024D44), Yunnan Province Xingdian Talent Support Program project funds.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to thank all the authors for their contributions.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eswa.2026.132319>.

Data availability

Data will be made available on request.

References

- Abbasi, K., Razzaghi, P., Gharizadeh, A., Ghareyazi, A., Dehnad, A., Rabiee, H. R., & Mofrad, M. R. (2025). Computational Drug Design in the AI Era: A Systematic Review of Molecular Representations, Generative Architectures, and Performance Assessment. *Pharmacological Reviews*, 100095.
- Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 2623–2631).
- Aminlari, A., East, M., Wei, W., & Quillen, D. (2008). Topiramate induced acute angle closure glaucoma. *The Open Ophthalmology Journal*, 2, 46.
- Bagherian, M., Sabeti, E., Wang, K., Sartor, M. A., Nikolovska-Coleska, Z., & Najarian, K. (2021). Machine learning approaches and databases for prediction of drug–target interaction: A survey paper. *Briefings in bioinformatics*, 22, 247–269.
- Ballester, P. J., & Mitchell, J. B. (2010). A machine learning approach to predicting protein–ligand binding affinity with applications to molecular docking. *Bioinformatics*, 26, 1169–1175.
- Bartholdi, D., Krajewska-Walasek, M., Öunap, K., Gaspar, H., Chrzanowska, K. H., Ilyana, H., Kayserili, H., Lurie, I. W., Schinzel, A., & Baumer, A. (2009). Epigenetic mutations of the imprinted IGF2-H19 domain in Silver–Russell syndrome (SRS): Results from a large cohort of patients with SRS and SRS-like phenotypes. *Journal of medical genetics*, 46, 192–197.
- Behr, B., Hoffmann, C., Ottolina, G., & Klotz, K.-N. (2006). Novel mutants of the human β_1 -adrenergic receptor reveal amino acids relevant for receptor activation. *Journal of Biological Chemistry*, 281, 18120–18125.
- Bigal, M. E., & Tepper, S. J. (2003). Ergotamine and dihydroergotamine: A review. *Current Pain and Headache Reports*, 7, 55–62.

- Bordes, A., Usunier, N., Garcia-Duran, A., Weston, J., & Yakhnenko, O. (2013). Translating embeddings for modeling multi-relational data. In *Advances in neural information processing systems* (p. 26).
- Cook, D., & Simons, D. J. (2023). Neuromuscular blockade. *StatPearls [Internet]*. StatPearls Publishing.
- Davis, A. P., Murphy, C. G., Johnson, R., Lay, J. M., Lennon-Hopkins, K., Saraceni-Richards, C., Sciaky, D., King, B. L., Rosenstein, M. C., Wieggers, T. C., & Mattingly, C. J. (2012). The Comparative Toxicogenomics Database: Update 2013. *Nucleic Acids Research*, 41, D1104–D1114.
- Dehghan, A., Razzaghi, P., Abbasi, K., & Gharaghani, S. (2023). TripletMultiDTI: Multimodal representation learning in drug-target interaction prediction with triplet loss function. *Expert Systems with Applications*, 232, Article 120754.
- Fang, J.-L., Wu, Y., Gamboa da Costa, G., Chen, S., Chitranshi, P., & Beland, F. A. (2016). Human sulfotransferases enhance the cytotoxicity of tolvaftan. *Toxicological sciences*, 150, 27–39.
- Gottlieb, A., Stein, G. Y., Rupp, E., & Sharan, R. (2011). PREDICT: A method for inferring novel drug indications with application to personalized medicine. *Molecular systems biology*, 7, 496.
- Grover, A., & Leskovec, J. (2016). node2vec: Scalable feature learning for networks. In *Proceedings of the 22nd ACM SIGKDD international conference on Knowledge discovery and data mining* (pp. 855–864).
- Gur, S., Yurdaarmagan, B., Bayatli, N., & Sikka, S. C. (2016). Effect of short-and long-term sildenafil treatment on erectile dysfunction in rats with partial bladder outlet obstruction. *Neurourology and Urodynamics*, 35, 108–114.
- Hansch, C., Maloney, P. P., Fujita, T., & Muir, R. M. (1962). Correlation of biological activity of phenoxyacetic acids with Hammett substituent constants and partition coefficients. *Nature*, 194, 178–180.
- He, T., Heidemeyer, M., Ban, F., Cherkasov, A., & Ester, M. (2017). SimBoost: A read-across approach for predicting drug–target binding affinities using gradient boosting machines. *Journal of cheminformatics*, 9, 24.
- Helbig, I., Meford, H. C., Sharp, A. J., Guipponi, M., Fichera, M., Franke, A., Muhle, H., De Kovel, C., Baker, C., & Von Spiczak, S. (2009). 15q13.3 microdeletions increase risk of idiopathic generalized epilepsy. *Nature genetics*, 41, 160–162.
- Huang, F., Qiu, Y., Li, Q., Liu, S., & Ni, F. (2020). Predicting drug-disease associations via multi-task learning based on collective matrix factorization. *Frontiers in Bioengineering and Biotechnology*, 8, 218.
- Huang, J., & Ling, C. X. (2005). Using AUC and accuracy in evaluating learning algorithms. *IEEE Transactions on Knowledge and Data Engineering*, 17, 299–310.
- Huang, Z., Kusan, M., Silva, A., & Singh, A. (2023). Link prediction without graph neural networks. *arXiv preprint arXiv:2305.13656*.
- Huang, Z., Silva, A., & Singh, A. (2021). A broader picture of random-walk based graph embedding. In *Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining* (pp. 685–695).
- Huang, Z., Xiao, Z., Ao, C., Guan, L., & Yu, L. (2025). Computational approaches for predicting drug-disease associations: A comprehensive review. *Frontiers of Computer Science*, 19, Article 195909.
- Keiser, M. J., Roth, B. L., Armbruster, B. N., Ernster, P., Irwin, J. J., & Shoichet, B. K. (2007). Relating protein pharmacology by ligand chemistry. *Nature biotechnology*, 25, 197–206.
- Kendall, A., Gal, Y., & Cipolla, R. (2018). Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7482–7491).
- Keshava Prasad, T., Goel, R., Kandasamy, K., Keerthikumar, S., Kumar, S., Mathivanan, S., Telikicherla, D., Raju, R., Shafreen, B., & Venugopal, A. (2009). Human protein reference database—2009 update. *Nucleic Acids Research*, 37, D767–D772.
- Kiranyaz, S., Ince, T., Abdeljaber, O., Avci, O., & Gabbouj, M. (2019). In *1-D convolutional neural networks for signal processing applications* (pp. 8360–8364). IEEE.
- Kitsirakou, S., Suratanee, A., & Plaimas, K. (2022). Integration of various protein similarities using random forest technique to infer augmented drug-protein matrix for enhancing drug-disease association prediction. *Science Progress*, 105, Article 00368504221109215.
- Knox, C., Law, V., Jewison, T., Liu, P., Ly, S., Frolkis, A., Pon, A., Banco, K., Mak, C., Neveu, V., Djoumbou, Y., Eisner, R., Guo, A. C., & Wishart, D. S. (2010). DrugBank 3.0: A comprehensive resource for 'omics' research on drugs. *Nucleic Acids Research*, 39, D1035–D1041.
- Kurz, M., Belani, K. G., Sessler, D. I., Kurz, A., Larson, M. D., Schroeder, M., & Blanchard, D. (1993). Naloxone, meperidine, and shivering. *Anesthesiology*, 79, 1193–1201.
- Leung, E., & Mitchelson, F. (1982). The interaction of pancuronium with cardiac and ileal muscarinic receptors. *European journal of pharmacology*, 80, 1–9.
- Li, Z., Huang, Q., Chen, X., Wang, Y., Li, J., Xie, Y., Dai, Z., & Zou, X. (2020). Identification of drug-disease associations using information of molecular structures and clinical symptoms via deep convolutional neural network. *Frontiers in Chemistry*, 7, 924.
- Lian, M., Wang, X., & Du, W. (2023). Drug-target interactions prediction based on network topology feature representation embedded deep forest. *Neurocomputing*, 551, Article 126509.
- Lin, X., Zhang, X., Yu, Z.-G., Long, Y., Zeng, X., & Yu, P. S. (2024). In *CSCL-DTI: predicting drug-target interaction through cross-view and self-supervised contrastive learning* (pp. 707–712). IEEE.
- Lü, L., & Zhou, T. (2011). Link prediction in complex networks: A survey. *Physica A: Statistical mechanics and its applications*, 390, 1150–1170.
- Lu, S., Liu, M., Liu, H., Yang, C., Zhu, J., Ling, Y., & Kuang, H. (2024). Gestational exposure to bisphenol AF causes endocrine disorder of corpus luteum by altering ovarian SIRT-1/Nrf2/NF-kB expressions and macrophage proangiogenic function in mice. *Biochemical Pharmacology*, 220, Article 115954.
- Luo, H., Li, M., Wang, S., Liu, Q., Li, Y., & Wang, J. (2018). Computational drug repositioning using low-rank matrix approximation and randomized algorithms. *Bioinformatics*, 34, 1904–1912.
- Luo, Y., Zhao, X., Zhou, J., Yang, J., Zhang, Y., Kuang, W., Peng, J., Chen, L., & Zeng, J. (2017). A network integration approach for drug-target interaction prediction and computational drug repositioning from heterogeneous information. *Nature communications*, 8, 573.
- Masoudi-Nejad, A., Mousavian, Z., & Bozorgmehr, J. H. (2013). Drug-target and disease networks: Polypharmacology in the post-genomic era. *In silico pharmacology*, 1, 17.
- Moreira, D. P., Ribeiro, Y. M., Ferreira, C. S., dos Santos Nassif Lacerda, S. M., & Rizzo, E. (2023). Exposure to acetaminophen impairs gametogenesis and fertility in zebrafish (*Danio rerio*). *Archives of Toxicology*, 97, 263–278.
- Morris, G. M., Huey, R., Lindstrom, W., Sanner, M. F., Belew, R. K., Goodsell, D. S., & Olson, A. J. (2009). AutoDock4 and AutoDockTools4: Automated docking with selective receptor flexibility. *Journal of computational chemistry*, 30, 2785–2791.
- Norouzi, R., Norouzi, R., Abbasi, K., Norouzi, R., & Razzaghi, P. (2025). DFT_ANPD: A dual-feature two-sided attention network for anticancer natural products detection. *Computers in Biology and Medicine*, 194, Article 110442.
- Oliva, P., Berrino, L., De Novellis, V., Palazzo, E., Marabese, I., Siniscalco, D., Scafuro, M., Mariani, L., Rossi, F., & Maione, S. (2006). Role of periaqueductal grey prostaglandin receptors in formalin-induced hyperalgesia. *European journal of pharmacology*, 530, 40–47.
- Öztürk, H., Özgür, A., & Ozkirimli, E. (2018). DeepDTA: Deep drug–target binding affinity prediction. *Bioinformatics*, 34, i821–i829.
- Peng, J., Li, J., & Shang, X. (2020). A learning-based method for drug-target interaction prediction based on feature representation learning and deep neural network. *BMC bioinformatics*, 21, 394.
- Peng, J., Wang, Y., Guan, J., Li, J., Han, R., Hao, J., Wei, Z., & Shang, X. (2021). An end-to-end heterogeneous graph representation learning-based framework for drug–target interaction prediction. *Briefings in bioinformatics*, 22, Article bbab430.
- Rc, g. (1998). Homologous up-regulation of vitamin D receptors is tissue specific in the rat. *J Bone Miner Res*, 13, 454–463.
- Rogers, S. W., Tvrdik, P., Capecchi, M. R., & Gahring, L. C. (2012). Prenatal ablation of nicotinic receptor alpha7 cell lineages produces lumbosacral spina bifida the severity of which is modified by choline and nicotine exposure. *American Journal of Medical Genetics Part A*, 158, 1135–1144.
- Sadeghi, S., Lu, J., & Ngom, A. (2022). A network-based drug repurposing method via non-negative matrix factorization. *Bioinformatics*, 38, 1369–1377.
- Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS one*, 10, Article e0118432.
- Santos, R., Ursu, O., Gaulton, A., Bento, A. P., Donadi, R. S., Bologa, C. G., Karlsson, A., Al-Lazikani, B., Hersey, A., & Oprea, T. I. (2017). A comprehensive map of molecular drug targets. *Nature reviews Drug discovery*, 16, 19–34.
- Schlichtkrull, M., Kipf, T. N., Bloem, P., Van Den Berg, R., Titov, I., & Welling, M. (2018). In *Modeling relational data with graph convolutional networks* (pp. 593–607). Springer.
- Shaikh, N., Sharma, M., & Garg, P. (2016). An improved approach for predicting drug–target interaction: Proteochemometrics to molecular docking. *Molecular Biosystems*, 12, 1006–1014.
- Shi, J.-Y., & Yiu, S.-M. (2015). In *SRP: A concise non-parametric similarity-rank-based model for predicting drug-target interactions* (pp. 1636–1641). IEEE.
- Shu, H., Hayashida, M., Arita, H., Huang, W., Zhang, H., An, K., Wu, G., & Hanaoka, K. (2011). Pentazocine-induced antinociception is mediated mainly by μ -opioid receptors and compromised by κ -opioid receptors in mice. *The Journal of pharmacology and experimental therapeutics*, 338, 579–587.
- Śledź, P., & Caffisch, A. (2018). Protein structure-based drug design: From docking to molecular dynamics. *Current opinion in structural biology*, 48, 93–102.
- Sohn, K. (2016). Improved deep metric learning with multi-class n-pair loss objective. In *Advances in neural information processing systems* (p. 29).
- Su, Y., Hu, Z., Wang, F., Bin, Y., Zheng, C., Li, H., Chen, H., & Zeng, X. (2024). AMGDTI: Drug–target interaction prediction based on adaptive meta-graph learning in heterogeneous network. *Briefings in bioinformatics*, 25.
- Tanvir, F., Saifuddin, K. M., Hossain, T., Bagavathi, A., & Akbas, E. (2023). HeTriNet: Heterogeneous graph triplet attention network for drug-target-disease interaction. *arXiv preprint arXiv:2312.00189*.
- Tian, A., Zhang, C., Rang, M., Yang, X., & Zhan, Z. (2020). RA-GCN: Relational aggregation graph convolutional network for knowledge graph completion. In *Proceedings of the 2020 12th international conference on machine learning and computing* (pp. 580–586).
- Tzoulis, P., Kaltsas, G., Baldeweg, S. E., Bouloux, P.-M., & Grossman, A. B. (2023). Tolvaftan for the treatment of the syndrome of inappropriate antidiuresis (SIAD). *Therapeutic Advances in Endocrinology and Metabolism*, 14, Article 20420188231173327.
- Wan, F., Hong, L., Xiao, A., Jiang, T., & Zeng, J. (2019). NeoDTI: Neural integration of neighbor information from a heterogeneous network for discovering new drug–target interactions. *Bioinformatics*, 35, 104–111.
- Wang, B., Zhang, Y., Ye, H., Wu, L., He, S., Bo, X., & Zhang, Z. (2023). In *Drug-target and Drug-disease Association Prediction based on Drug-target-disease Network and Multi-task Learning* (pp. 930–935). IEEE.
- Wang, G., Zhang, H., Shao, M., Sun, S., & Cao, C. (2025). MMPD-DTA: Integrating multi-modal deep learning with pocket-drug graphs for drug-target binding affinity prediction. *Journal of Chemical Information and Modeling*, 65, 1615–1630.

- Xia, Z., Wu, L.-Y., Zhou, X., & Wong, S. T. (2010). Semi-supervised drug-protein interaction prediction from heterogeneous biological spaces. *BMC systems biology*, 4, S6.
- Xie, C., Hu, J., Motloch, L. J., Karam, B. S., & Akar, F. G. (2015). The Classically Cardioprotective Agent Diazoxide Elicits Arrhythmias in Type 2 Diabetes Mellitus. *J Am Coll Cardiol*, 66, 1144–1156.
- Xu, J., Li, Z., Du, B., Zhang, M., & Liu, J. (2020). In *Reluplex made more practical: Leaky ReLU* (pp. 1–7). IEEE.
- Xu, Y.-C., Schaus, J. M., Walker, C., Krushinski, J., Adham, N., Zgombick, J. M., Liang, S. X., Kohlman, D. T., & Audia, J. E. (1999). N-Methyl-5-tert-butyltryptamine: A novel, highly potent 5-HT_{1D} receptor agonist. *Journal of medicinal chemistry*, 42, 526–531.
- Xuan, P., Ye, Y., Zhang, T., Zhao, L., & Sun, C. (2019). Convolutional neural network and bidirectional long short-term memory-based method for predicting drug–disease associations. *Cells*, 8, 705.
- Yang, B., Yih, W.-t., He, X., Gao, J., & Deng, L. (2014). Embedding entities and relations for learning and inference in knowledge bases. *arXiv preprint arXiv:1412.6575*.
- Yang, K., Zhao, X., Waxman, D., & Zhao, X.-M. (2019). Predicting drug-disease associations with heterogeneous network embedding. *Chaos: An Interdisciplinary. Journal of Nonlinear Science*, 29.
- Yang, M., Luo, H., Li, Y., & Wang, J. (2019). Drug repositioning based on bounded nuclear norm regularization. *Bioinformatics*, 35, i455–i463.
- Yang, M., Wu, G., Zhao, Q., Li, Y., & Wang, J. (2021). Computational drug repositioning based on multi-similarities bilinear matrix factorization. *Briefings in bioinformatics*, 22, Article bbab267.
- Yang, R., Fu, Y., Zhang, Q., & Zhang, L. (2024). GCNGAT: Drug–disease association prediction based on graph convolution neural network and graph attention network. *Artificial Intelligence in Medicine*, 150, Article 102805.
- Yu, Z., Huang, F., Zhao, X., Xiao, W., & Zhang, W. (2021). Predicting drug–disease associations through layer attention graph convolutional network. *Briefings in bioinformatics*, 22, Article bbab243.
- Zeng, X., Zhu, S., Liu, X., Zhou, Y., Nussinov, R., & Cheng, F. (2019). deepDR: A network-based deep learning approach to in silico drug repositioning. *Bioinformatics*, 35, 5191–5198.
- Zhang, W., Klimek, V., Farley, J. T., Zhu, M.-Y., & Ordway, G. A. (1999). α 2C adrenoceptors inhibit adenylyl cyclase in mouse striatum: Potential activation by dopamine. *The Journal of pharmacology and experimental therapeutics*, 289, 1286–1292.
- Zhao, B.-W., Hu, L., You, Z.-H., Wang, L., & Su, X.-R. (2022). HINGRL: Predicting drug–disease associations with graph representation learning on heterogeneous information networks. *Briefings in bioinformatics*, 23, Article bbab515.